

PRAKTIKUM BASIS DATA

MODUL 11

NoSQL_Key_Value_Database



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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UNIVERSITAS TELKOM

2026

JURNAL

1. A)

```
Connected to Test Cluster at 127.0.0.1:9042.
[cqlsh 5.0.1 | Cassandra 2.2.3 | CQL spec 3.3.1 | Native protocol v4]
Use HELP for help.
WARNING: pyreadline dependency missing. Install to enable tab completion.
cqlsh> CREATE KEYSPACE spotifine_0802
... WITH replication = {'class':'SimpleStrategy','replication_factor':1};
cqlsh> USE spotifine_0802;
cqlsh:spotifine_0802> |
```

B)

```
Cassandra CQL Shell
Connected to Test Cluster at 127.0.0.1:9042.
[cqlsh 5.0.1 | Cassandra 2.2.3 | CQL spec 3.3.1 | Native protocol v4]
Use HELP for help.
WARNING: pyreadline dependency missing. Install to enable tab completion.
cqlsh> CREATE KEYSPACE spotifine_0802
... WITH replication = {'class':'SimpleStrategy','replication_factor':1};
cqlsh> USE spotifine_0802;
cqlsh:spotifine_0802> CREATE TABLE musician_data(
...     artist_id UUID PRIMARY KEY,
...     artist_name TEXT,
...     followers INT,
...     origin TEXT,
...     genres SET<TEXT>
... );
cqlsh:spotifine_0802> |
```

```
Cassandra CQL Shell
Connected to Test Cluster at 127.0.0.1:9042.
[cqlsh 5.0.1 | Cassandra 2.2.3 | CQL spec 3.3.1 | Native protocol v4]
Use HELP for help.
WARNING: pyreadline dependency missing. Install to enable tab completion.
cqlsh> CREATE KEYSPACE spotifine_0802
... WITH replication = {'class': 'SimpleStrategy', 'replication_factor':1};
cqlsh> USE spotifine_0802;
cqlsh:spotifine_0802> CREATE TABLE musician_data(
... artist_id UUID PRIMARY KEY,
... artist_name TEXT,
... followers INT,
... origin TEXT,
... genres SET<TEXT>
... );
cqlsh:spotifine_0802> INSERT INTO musician_data(artist_id,artist_name,followers,origin,genres)
... VALUES(uuid(), 'Dave Rodgers', 5000000, 'Italy', {'Eurobeat', 'Dance'});
cqlsh:spotifine_0802> INSERT INTO musician_data(artist_id,artist_name,followers,origin,genres)
... VALUES(uuid(), 'YOASOBI', 25000000, 'Japan', {'J-Pop', 'Pop'});
cqlsh:spotifine_0802> INSERT INTO musician_data(artist_id,artist_name,followers,origin,genres)
... VALUES(uuid(), 'Manuel', 3000000, 'Italy', {'Eurobeat'});
cqlsh:spotifine_0802>
```

```
Cassandra CQL Shell
Connected to Test Cluster at 127.0.0.1:9042.
[cqlsh 5.0.1 | Cassandra 2.2.3 | CQL spec 3.3.1 | Native protocol v4]
Use HELP for help.
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cqlsh> CREATE KEYSPACE spotifine_0802
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cqlsh> USE spotifine_0802;
cqlsh:spotifine_0802> CREATE TABLE musician_data(
... artist_id UUID PRIMARY KEY,
... artist_name TEXT,
... followers INT,
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cqlsh:spotifine_0802> INSERT INTO musician_data(artist_id,artist_name,followers,origin,genres)
... VALUES(uuid(), 'Dave Rodgers', 5000000, 'Italy', {'Eurobeat', 'Dance'});
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... VALUES(uuid(), 'YOASOBI', 25000000, 'Japan', {'J-Pop', 'Pop'});
cqlsh:spotifine_0802> INSERT INTO musician_data(artist_id,artist_name,followers,origin,genres)
... VALUES(uuid(), 'Manuel', 3000000, 'Italy', {'Eurobeat'});
cqlsh:spotifine_0802> SELECT * FROM musician_data;

artist_id | artist_name | followers | genres | origin
-----+-----+-----+-----+-----
b5ee5e90-b989-4396-b526-12a661274ffe | Dave Rodgers | 5000000 | {'Dance', 'Eurobeat'} | Italy
e93fc15a-4a80-4dd6-9083-e71d8c59dbdd | YOASOBI | 25000000 | {'J-Pop', 'Pop'} | Japan
1ab26654-2054-437c-97a8-208919485768 | Manuel | 3000000 | {'Eurobeat'} | Italy

(3 rows)
cqlsh:spotifine_0802> |
```

2. Apa keuntungan utama menggunakan tipe data set pada kolom genres di aplikasi Spotifine dibandingkan jika ia menggunakan tabel relasional (RDBMS) biasa?

Keuntungan utamanya itu karena menggunakan tipe data set pada kolom genres di Cassandra karena satu musisi bisa memiliki lebih dari satu genre tanpa perlu membuat tabel tambahan seperti pada database relasional biasa. Dengan set, semua genre dapat langsung disimpan dalam satu kolom sehingga struktur database menjadi lebih sederhana dan proses pengambilan data menjadi lebih cepat. Selain itu, tipe data set juga otomatis mencegah data duplikat. Jika genre yang sama dimasukkan lebih dari satu kali, Cassandra hanya akan menyimpan satu data saja. Hal ini membuat penyimpanan data menjadi lebih efisien dan cocok digunakan untuk aplikasi besar seperti Spotifine.